ORF409 HW5 (Ch8)

(Q.1) $\sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum_{i=1}^{n}(x_i^2-2x_i\bar{x}+\bar{x}^2)$

$$= \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} 2x_i\bar{x} + \sum_{i=1}^{n}\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - 2\bar{x}\sum_{i=1}^{n} x_i + n\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - 2n\bar{x}^2 + n\bar{x}^2$$

$$= \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

— QED

(Q.2) Let X = R.V taking values $x_1, \ldots, x_n$

Random Variable

So, $\cancel{\sum_{i=1}^{n}}$ $\text{var}(X) = E(X^2) - (E(X))^2$ → Here $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$

$$\Rightarrow \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)^2$$

$$\Rightarrow \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \bar{x}^2$$

$$\Rightarrow \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

- QED

ORF 409 HW5 ~~Rough~~

(Q.4)
(a) Note that if $X_1, X_2, \ldots, X_n$ are iid std normal samples

$N(\mu, \sigma^2)$, $\mu \to 0$, $\sigma^2 \to 1$

& $\bar{X} = \frac{\sum X_i}{n}$,

$$\frac{\bar{X} - E(\bar{X})}{\sqrt{Var(\bar{X})}} \Rightarrow N(0,1)$$

$$\Rightarrow \frac{\bar{X} - E(\bar{X})}{\sqrt{\frac{1}{n^2} \cdot n \, Var(X_i)}} \Rightarrow N(0,1)$$

$Var(X_i) \to \sim s^2$

So, $\frac{s}{\sqrt{n}} < 0.1$

$\Rightarrow \frac{\sigma}{\sqrt{n}} < 0.1$ ($\because$ $s^2 (= \frac{1}{n-1} \sum (x_i - \bar{x})^2)$ is an unbiased estimator of $var(X_i) = \sigma^2$, $s \xrightarrow{w.ar^m} \sigma$)

$\Rightarrow \frac{1}{\sqrt{n}} < 0.1$

$\Rightarrow \sqrt{n} > 10$

$\Rightarrow n > 100$

~~$\Rightarrow$ (We expect 100 numbers)~~

$\Rightarrow$ we expect ~~101 numbers~~ $\sim > 100$ numbers

~ 100 numbers (101, 102 maybe)

Q. 9)

(a) $P(M>n) = P(U_1 \le U_2 \le \text{—} \le U_n)$

~~n! ways to arrange $[U_1, U_2, \text{—}, U_n]$ each equally likely~~

$[U_1, U_2, \text{—}, U_n]$ are n iid $U[0,1]$ draws.

Since they are (independent) & (identically distributed), each of the n! possibilities (of arranging these from smallest to largest) is equally likely.

$n! \Rightarrow [U_1, U_2, \text{—}, U_{n-1}, U_n] \rightarrow$ This means: $U_1 \le U_2 \le \text{—} \le U_{n-1} \le U_n$

$U_1, U_2, \text{—}, U_n, U_{n-1}]$

$|$

$[U_n, U_{n-1}, \text{—}, U_2, U_1]$

$\Rightarrow P([U_1, U_2, \text{—}, U_{n-1}, U_n]) = \frac{1}{n!}$

$\Rightarrow P(U_1 \le U_2 \le \text{—} \le U_{n-1} \le U_n) = \frac{1}{n!}$

$\Rightarrow P(M>n) = \frac{1}{n!}$

- QED.

(b) $E(M) = \sum_{n=0}^{\infty} P(M>n)$

$= \sum_{n=0}^{\infty} \frac{1}{n!}$

$= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \dots$

$= e^1$ ~~(by Taylor expansion~~

$= e$ (By Taylor series expansion:

$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$\Rightarrow e^1 = \sum_{n=0}^{\infty} \frac{1^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!}$)

(Q.16)

M2: Samples $\rightarrow$ d_1 d_2 ——— d_k

n_1 n_2 n_k

M2:

$$\text{mean} = \frac{1}{k}\left(\frac{d_1}{n_1} + \frac{d_2}{n_2} + \cdots + \frac{d_k}{n_k}\right)$$

$$= \frac{d_1}{n_1}\left(\frac{1}{k}\right) + \cdots + \frac{d_k}{n_k}\left(\frac{1}{k}\right)$$

wts: equal to each cycle but NOT for each person

same for

C1 person $\Rightarrow$ wt $= \frac{1}{n_1 k}$

C_2 — || — $\frac{1}{n_2 k}$

M3:

$$\text{mean} = \frac{d_1 + d_2 + \cdots + d_k}{n_1 + n_2 + \cdots + n_k}$$

$$= \frac{d_1}{n_1}\left(\frac{n_1}{\Sigma n_i}\right) + \cdots + \frac{d_k}{n_k} \cdot \frac{n_k}{(\Sigma n_i)}$$

same wt for each person but not for each cycle

$\Rightarrow$ cycle$_i$ wt $\propto n_i$

since we want avg time "a person" spends in the system, giving equal wt to each person makes more sense.

$\Rightarrow$ M3 ✓

Now, MSE $= E_F(g(\bar{x}) - \theta(F))^2$

$$= E(\bar{x} - E(x))^2$$

$$= E(\bar{x} - E(\bar{x}))^2$$

$$= Var(\bar{x})$$

$$= \frac{Var(x_i)}{n}$$

$$= \frac{\sigma^2}{n}$$

$$\approx \frac{s^2}{n}$$

$$= \frac{\Sigma(x_i - \bar{x})^2 / n-1}{n} = \frac{\Sigma(x_i - \bar{x})^2}{n(n-1)}$$

It depends on what you are trying to estimate (variable of interest).

If it is $\frac{D_i}{N_i}$,

Σd_i, Σn_i

d_i, n_i

Our study is finding expected value of avg time; not just the avg time.

If it were avg time by each person

1 system simulation $\Rightarrow$ $\frac{\text{Total Time}}{N}$ $\Rightarrow$ avg time by a person

But $\frac{D_i}{N_i}$ represents avg time variable.

say $\hookrightarrow Y_i$ (D_i = Total time in i^{th} simlⁿ,
N_i = -||- ppl — || —)

In the book's eg,

1 run $\rightarrow$ $\frac{\Sigma D_i}{\Sigma N_i}$ represents ? ((realized value of avg time?

If so, we just computed var, Not MSE of (Avg time)